The z integral of $d^3N_{2b\,\mathrm{UV}}/d^3k$ is done exactly, in $d_\perp = 2-\epsilon$ ($\overline{MS}$, $\bar\mu^2 = 4\pi e^{-\gamma_E}\mu^2$). It uses a Feynman parameter α and the Fourier transform $\int d^dl\, e^{iq\cdot l}(l^2+\Delta)^{-n}$. K_0, K_1, K_2 are modified Bessel functions; all of them have the same argument $|k|\,|y-x|\sqrt{\alpha(1-\xi)\big(1-\alpha(1-\xi)\big)}$, written out in full below. Here $|k| = \sqrt{k^2}$ and $|y-x| = \sqrt{(y-x)^2}$ (transverse).

$$
\begin{aligned}
\frac{d^3N_{2b\,\mathrm{UV}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{\Lambda/k^+}^{1-\Lambda/k^+}\frac{d\xi}{2\pi}\int_{x,y,x',w'} e^{-ik\cdot(w'-x)}\Bigg\{
\\
\pi\,\frac{(x'-w')\cdot(y-x)}{(x'-w')^2\,(y-x)^2}\left[\Big(\xi(1-\xi)+\frac{\xi}{1-\xi}+\frac{1-\xi}{\xi}\Big)\Big(\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y-x)^2\,\bar\mu^2}{4\,(1-\xi)}\Big)+\xi(1-\xi)\right]
\\
+\Big(\xi(1-\xi)+\frac{\xi}{1-\xi}+\frac{1-\xi}{\xi}\Big)\,\pi\,\frac{(x'-w')\cdot(y-x)}{(x'-w')^2}
\\
\times\int_0^1\frac{d\alpha}{\alpha}\left[\frac{|k|\sqrt{\alpha(1-\xi)}\;e^{i\alpha(1-\xi)\,k\cdot(y-x)}}{|y-x|\sqrt{1-\alpha(1-\xi)}}\,K_1\Big(|k|\,|y-x|\sqrt{\alpha(1-\xi)\big(1-\alpha(1-\xi)\big)}\Big)-\frac{1}{(y-x)^2}\right]
\\
+\frac{2\,\xi(1-\xi)}{(x'-w')^2}\int_0^1 d\alpha\; e^{i\alpha(1-\xi)\,k\cdot(y-x)}\Bigg\{
\\
\frac{\pi(1-\alpha)}{2}\Bigg[\frac{\sqrt{1-\xi}\,|k|}{|y-x|\sqrt{\alpha\big(1-\alpha(1-\xi)\big)}}\Bigg((x'-w')\cdot(y-x)\,K_1\Big(|k|\,|y-x|\sqrt{\alpha(1-\xi)\big(1-\alpha(1-\xi)\big)}\Big)
\\
-\frac{\big(k\cdot(x'-w')\big)\big(k\cdot(y-x)\big)}{k^2}\,|k|\,|y-x|\sqrt{\alpha(1-\xi)\big(1-\alpha(1-\xi)\big)}\;K_0\Big(|k|\,|y-x|\sqrt{\alpha(1-\xi)\big(1-\alpha(1-\xi)\big)}\Big)\Bigg)
\\
+\frac{i\,\alpha\,(1-\xi)^{3/2}\,|k|}{|y-x|\sqrt{\alpha\big(1-\alpha(1-\xi)\big)}}\Big((x'-w')\cdot(y-x)\;k\cdot(y-x)+(y-x)^2\;k\cdot(x'-w')\Big)
\\
\times K_1\Big(|k|\,|y-x|\sqrt{\alpha(1-\xi)\big(1-\alpha(1-\xi)\big)}\Big)\Big)
\\
+\frac{\alpha\,(1-\xi)^2\,k^2\,(x'-w')\cdot(y-x)}{1-\alpha(1-\xi)}\,K_2\Big(|k|\,|y-x|\sqrt{\alpha(1-\xi)\big(1-\alpha(1-\xi)\big)}\Big)\Bigg]
\\
-\frac{\pi\,(x'-w')\cdot(y-x)\,|k|\sqrt{1-\xi}}{2\,|y-x|\sqrt{\alpha\big(1-\alpha(1-\xi)\big)}}\,K_1\Big(|k|\,|y-x|\sqrt{\alpha(1-\xi)\big(1-\alpha(1-\xi)\big)}\Big)\Bigg\}\Bigg\}
\\
\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(x)\Big]\rho^{b'}(x')\,\rho^{f}(y).
\end{aligned}
\tag{1}
$$

Remarks

- In the convention $\int d^2p/p^2 = 0$ (all $1/\epsilon$ dropped), delete the $\frac{2}{\epsilon}$ in the first line. Nothing else changes.
- The only $\bar\mu$ dependence is $\log\bar\mu^2$ in the first line. After the ξ integral, $\int_{\Lambda/k^+}^{1-\Lambda/k^+} d\xi\,\big(\xi(1-\xi)+\frac{\xi}{1-\xi}+\frac{1-\xi}{\xi}\big) = 2\log\frac{k^+-\Lambda}{\Lambda} - \frac{11}{6} + O(\Lambda/k^+)$, so its coefficient is $\frac{1}{(2\pi)^3}\frac{g^4}{16\pi^4}\frac{1}{k^+}\big[2\log\frac{k^+}{\Lambda}-\frac{11}{6}\big]$.
- The second and third lines are finite. Both α integrals converge at $\alpha\to 0$, and their integrands behave as $\alpha^{-1/2}$ or better.
- The $-\frac{1}{(y-x)^2}$ inside the second line and the $+\xi(1-\xi)$ in the first line come from the ϵ-expansion of the d-dimensional Feynman-parameter integral; they are not added by hand.
- The two pieces were checked numerically against a direct two-dimensional z integration with a small-$|z-x|$ cutoff δ (the cutoff replaces $\frac{2}{\epsilon}+2\gamma_E+\log\frac{\bar\mu^2}{4}$ by $\log\frac{1}{\delta^2}$). They agree to 5×10^{-5}, which is the accuracy of the grid.
- The $\log\frac{1}{1-\xi}$ in the first line gives $\frac{1}{2}\log^2\frac{k^+}{\Lambda}$ after the ξ integral. This is the large-$|z|$ (infrared) behaviour at $\xi\to1$, and it has to cancel against the rest of N_{2b} and the other JIMWLK-type terms.